Cambridge Instruments & Engg. Co.

EST. 1963 · HOWRAH, INDIA

DIGITAL MULTIMETER · TRUE-RMS, MANUAL RANGING

# DM 321T

True-RMS · 3½-digit · manual ranging · hFE · capacitance · backlit

## 01 KEY FEATURES

- True-RMS AC measurement
- 3½-digit backlit display
- Data hold
- Auto-ranging capacitance
- hFE transistor test
- NCV detection and built-in torch



DM 321T · DIGITAL MULTIMETER

## 02 TECHNICAL SPECIFICATIONS

| PARAMETER | SPECIFICATION & ACCURACY |
|---|---|
| **DISPLAY & RANGING** | |
| Measurement Type | True-RMS |
| Ranging | Manual |
| Display | 3½-digit LCD with backlight, maximum reading 1999 |
| **VOLTAGE & CURRENT** | |
| DC Voltage | 200 mV / 2 V / 20 V / 200 V / 1000 V \| ±(0.5% + 3) on 200 mV–200 V, ±(0.8% + 10) on 1000 V |
| AC Voltage (True-RMS) | 2 V / 20 V / 200 V / 750 V \| ±(0.8% + 5) on 2–200 V, ±(1.2% + 10) on 750 V \| 40 Hz–1 kHz |
| DC Current | 200 µA / 2 mA / 20 mA / 200 mA / 20 A \| ±(0.8% + 10) on 200 µA–20 mA, ±(1.2% + 8) on 200 mA, ±(2.0% + 5) on 20 A |
| AC Current (True-RMS) | 20 mA / 200 mA / 20 A \| ±(1.0% + 15) on 20 mA, ±(2.0% + 5) on 200 mA, ±(3.0% + 10) on 20 A \| 40 Hz–400 Hz |
| **OTHER MEASUREMENTS** | |
| Resistance | 200Ω / 2KΩ / 20KΩ / 200KΩ / 2MΩ / 20MΩ \| ±(0.8% + 5) on 200 Ω, ±(0.8% + 3) on 2KΩ–2MΩ, ±(1.0% + 25) on 20MΩ |
| Capacitance (Auto Ranging) | 6 nF – 20 mF \| ±(5.0% + 40) on 6 nF, ±(3.5% + 20) on 60 nF–60 µF, ±(5.0% + 10) on 600 µF–20 mF |
| Transistor Test | hFE, NPN and PNP, 0–1000 |
| Additional Functions | Continuity buzzer (below approx. 50 ± 20 Ω), diode test, REL, AC mains frequency test (750 V range), NCV detection (48–220 V AC), torch, data hold, backlight, auto power-off, low battery |
| **GENERAL** | |
| Power Supply | 2 × 1.5 V AA battery |
| Fuses | mA input 200 mA / 250 V, 20 A input 20 A / 250 V · 20 A max 10 s |
| Operating Temperature | 0 °C to 40 °C, below 75% RH |
| Dimensions (L×W×H) | 178 × 86 × 52 mm |
| Weight | Approx. 335 g (incl. battery), 360 g with holster |
| Warranty | 12 months against manufacturing defects only |

## 03 APPLICATIONS

- General electrical maintenance and panel testing
- True-RMS AC on non-sinusoidal loads
- Basic electronic troubleshooting incl. transistor testing

## 04 WHAT'S IN THE BOX

✓ Test leads ✓ User manual ✓ Battery ✓ Holster



All specifications are subject to change without prior notice. © Cambridge Instruments & Engg. Co. - DM 321T Product Catalogue.